Monthly Consumables Management System – Documentation

1. Overview

This project automates the management and reporting of IT consumables using Microsoft SharePoint, Power Apps, and Power Automate. It covers consumables stock tracking, release logging, and automated monthly email reports.

2. Components

- SharePoint Lists
  - Consumables
  - Consumables_Release
- Power Apps (Data entry and user interaction)
- Power Automate (Scheduled reporting and email notifications)
- Outlook (Email delivery)

3. SharePoint Lists

3.1 Consumables List

Purpose: Stores master data for consumable items and stock levels.

Main Columns:
- Item Name
- Oracle Code
- Stock
- No of Printers
- To Order
- Site

3.2 Consumables_Release List

Purpose: Logs each consumable release transaction.

Main Columns:
- Release Date (Date column)
- Item (Lookup to Consumables)
- Oracle Code
- Quantity
- Printer
- Released To (User)
- Location
- Released By (IT Member)
- Comment

4. Power Apps

Power Apps is used as the front-end for releasing consumables.

Key Features:
- User selection using Office365Users.SearchUser
- Controlled data entry for quantity, printer, and location
- Submission writes data to Consumables_Release list
- Stock calculations are handled through Power Automate / list logic

5. Power Automate Flows

5.1 Monthly Consumables Release Report

Trigger:
- Recurrence (Monthly)

Flow Steps:
1. Compose – StartOfLastMonth
2. Compose – EndOfLastMonth
3. Get items (Consumables_Release)
   - Filter by Release Date between start and end of last month
4. Select
   - Map required columns only
5. Create HTML table
6. Compose – StyledTable
   - Adds CSS styling and borders to the HTML table
7. Send an email (V2)
   - Sends formatted report via Outlook

5.2 Monthly Consumables Stock Status Report

Trigger:
- Recurrence (Monthly)

Flow Steps:
1. Get items (Consumables)
2. Select (prepare stock data)
3. Create HTML table
4. StyledTable (CSS borders)
5. Send an email (V2)

6. HTML Table Styling

Custom CSS is applied to all email tables using a Compose action:

- Border collapse enabled
- Table width set to 100%
- Header background color
- Cell borders and padding

This ensures consistent and professional email formatting.

7. Common Issues & Solutions

- Extra {"body": ""} appearing in emails:
  - Caused by mixing outputs('Create_HTML_table') and body('Create_HTML_table')
  - Solution: Use only outputs('StyledTable') in email body

- Item or user fields appearing empty during edit:
  - Caused by incorrect Power Apps card Update property
  - Solution: Ensure Update uses Selected/Value correctly

8. Security & Permissions

- Users have limited access to Power Apps
- Only IT team can edit SharePoint lists
- Emails sent using service account (Power Apps / IT Infrastructure)

9. Future Enhancements

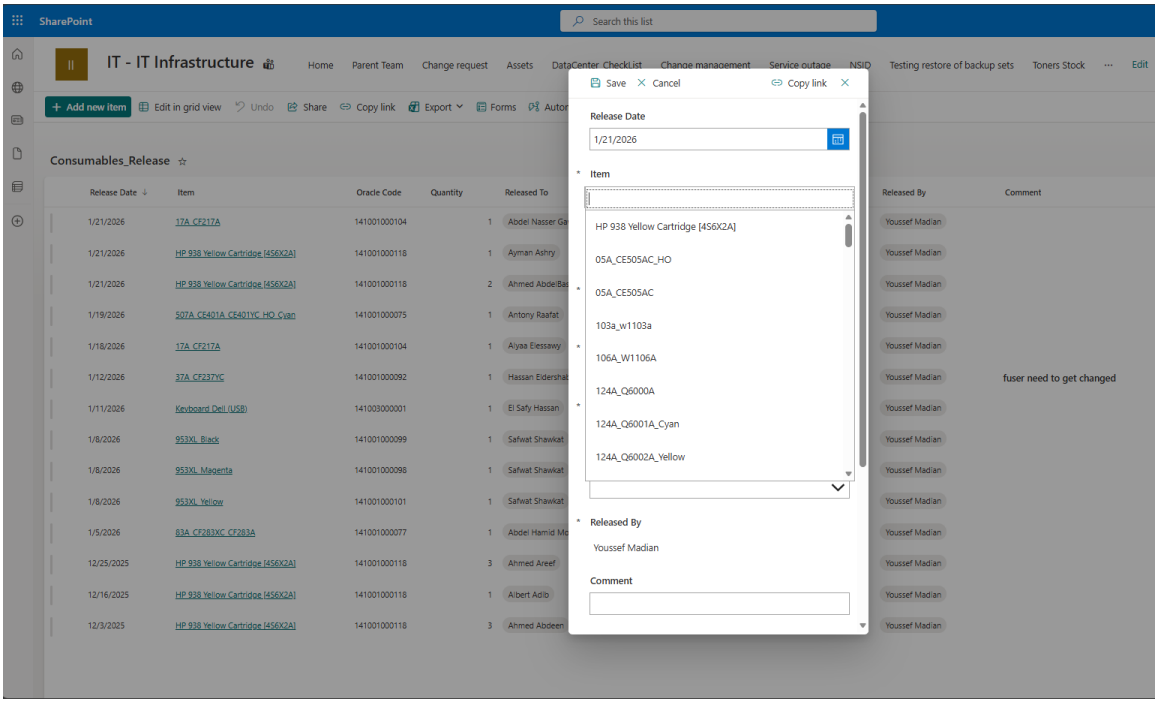
- Dashboard using Power BI
- Low-stock alerts
- Approval workflow for large quantity releases
- Multi-site reporting

Operational Workflow (How the System Works)

This section explains the end-to-end operational workflow of the Consumables Management System.

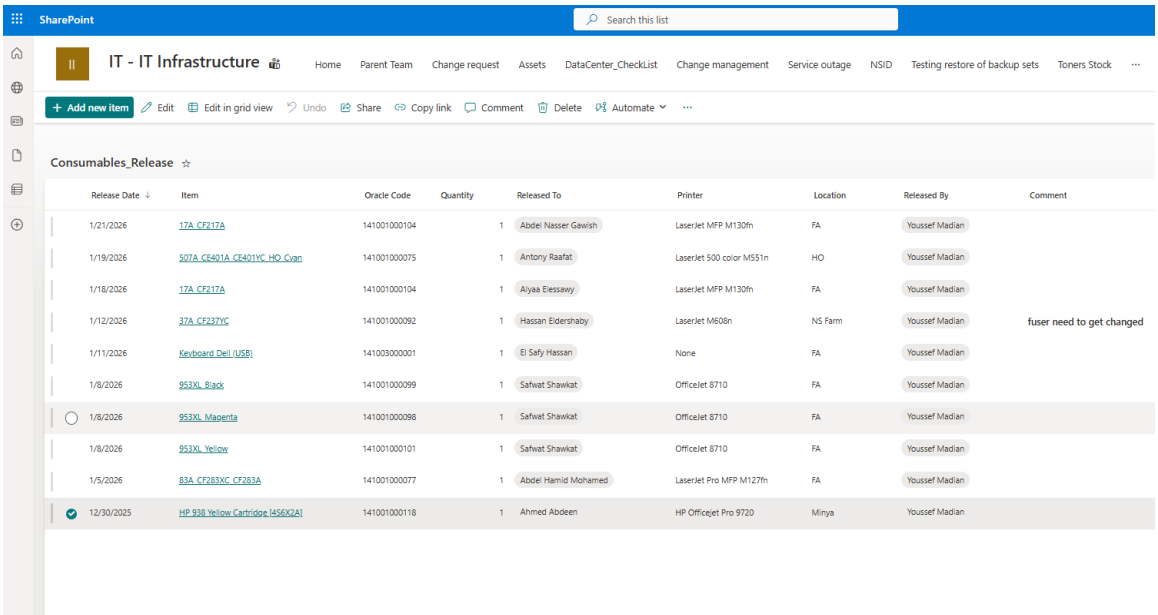
1. User Interaction (Power Apps)

The user opens the Consumables Power App and selects the required item, quantity, printer, and released-to user. Mandatory fields are validated before submission. The quantity will then be automatically deducted from stock in the Consumables list.



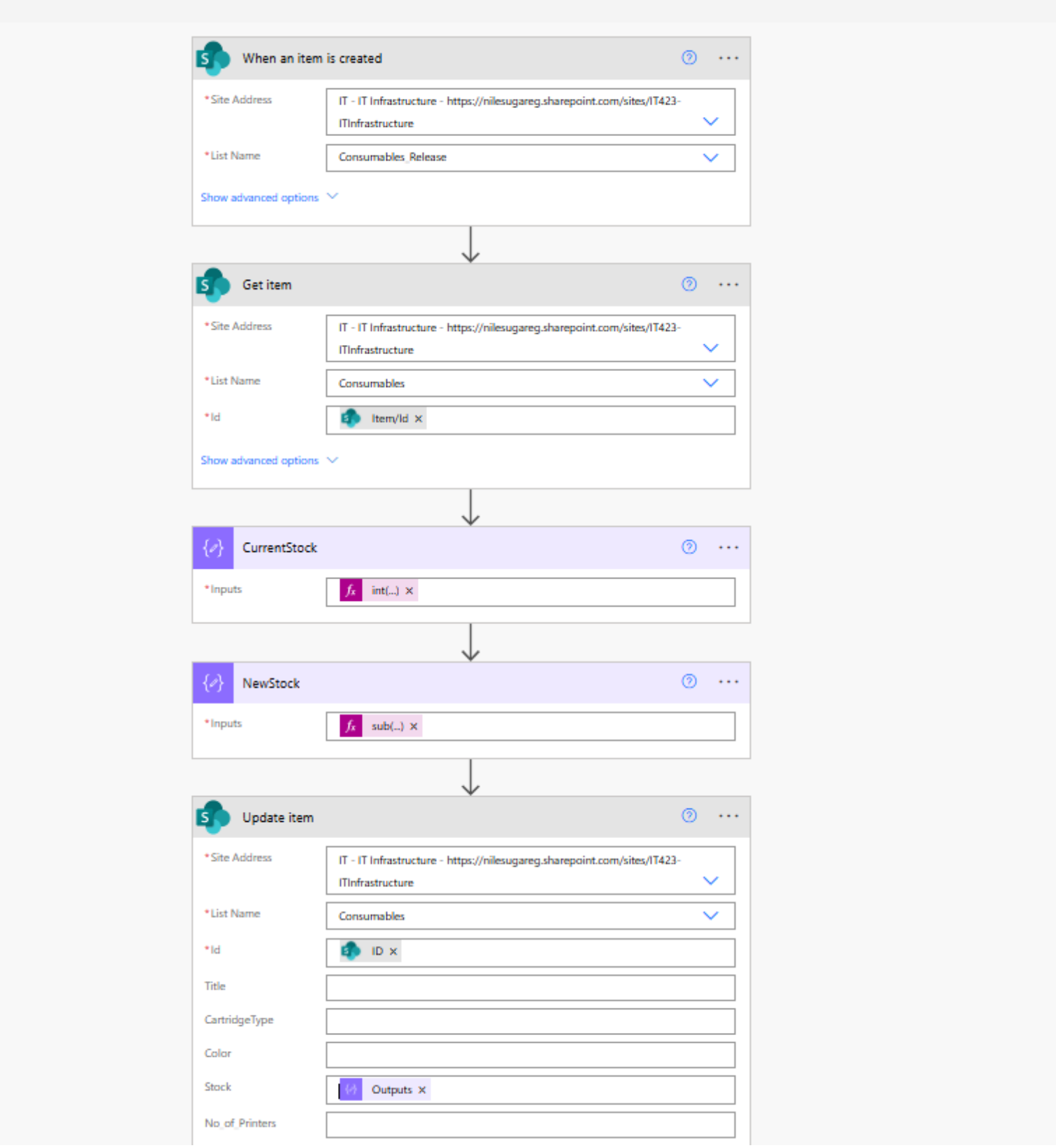
2. Release Transaction Saved

Once the user submits the form, a new record is created in the SharePoint list 'Consumables_Release'. This list acts as the transaction log for all releases.



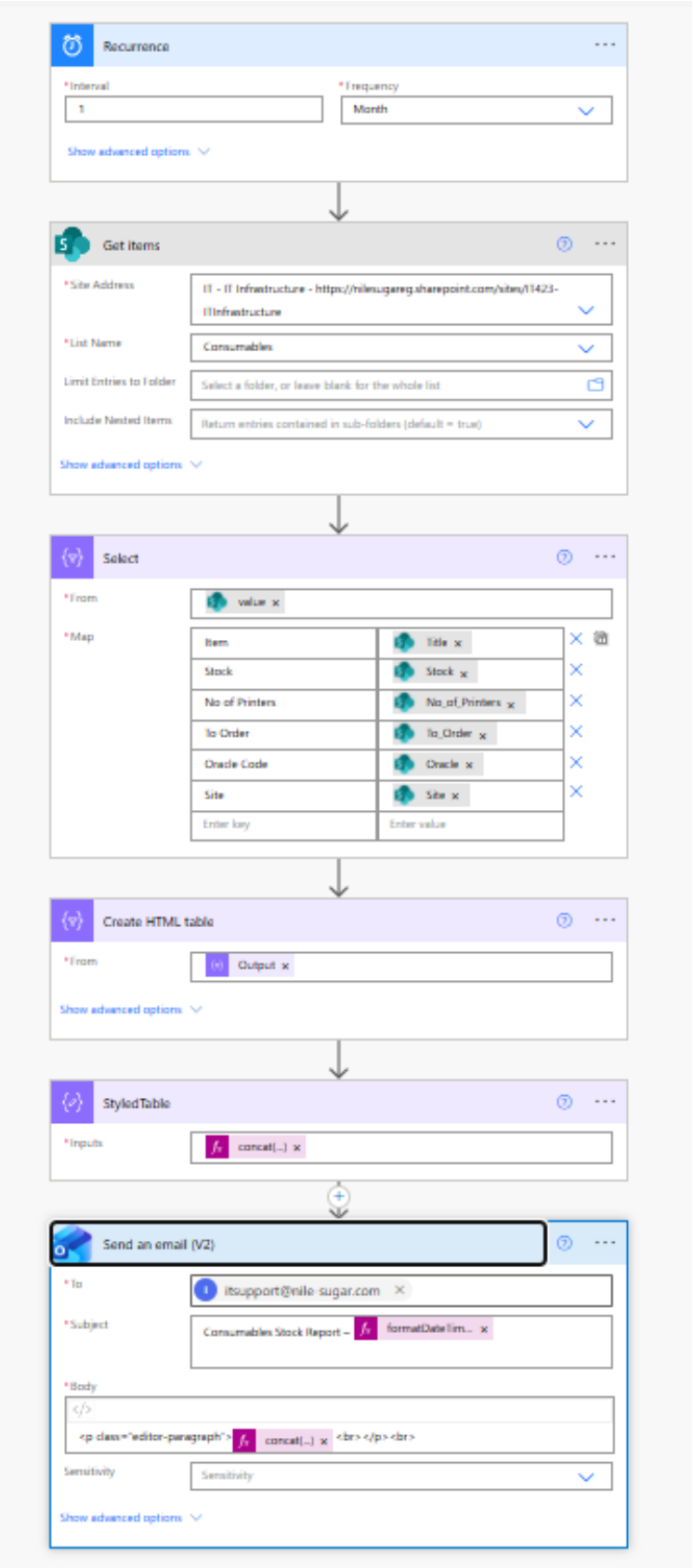
3. Stock Update Logic

A Power Automate flow is triggered to update the stock in the 'Consumables' list. The flow retrieves the current stock value, subtracts the released quantity, and updates the remaining stock correctly.



4. Monthly Stock Status Email

On the 1st day of every month, a scheduled Power Automate flow retrieves all consumables stock data. The data is formatted into an HTML table with borders and sent via email to stakeholders.



Monthly Stock Status Email Template:

Consumables Stock Report – January 2026

Power_Apps
To: Youssef Madian

Reply | Reply all | Forward

Wed 2026-01-21 09:10

Dears,

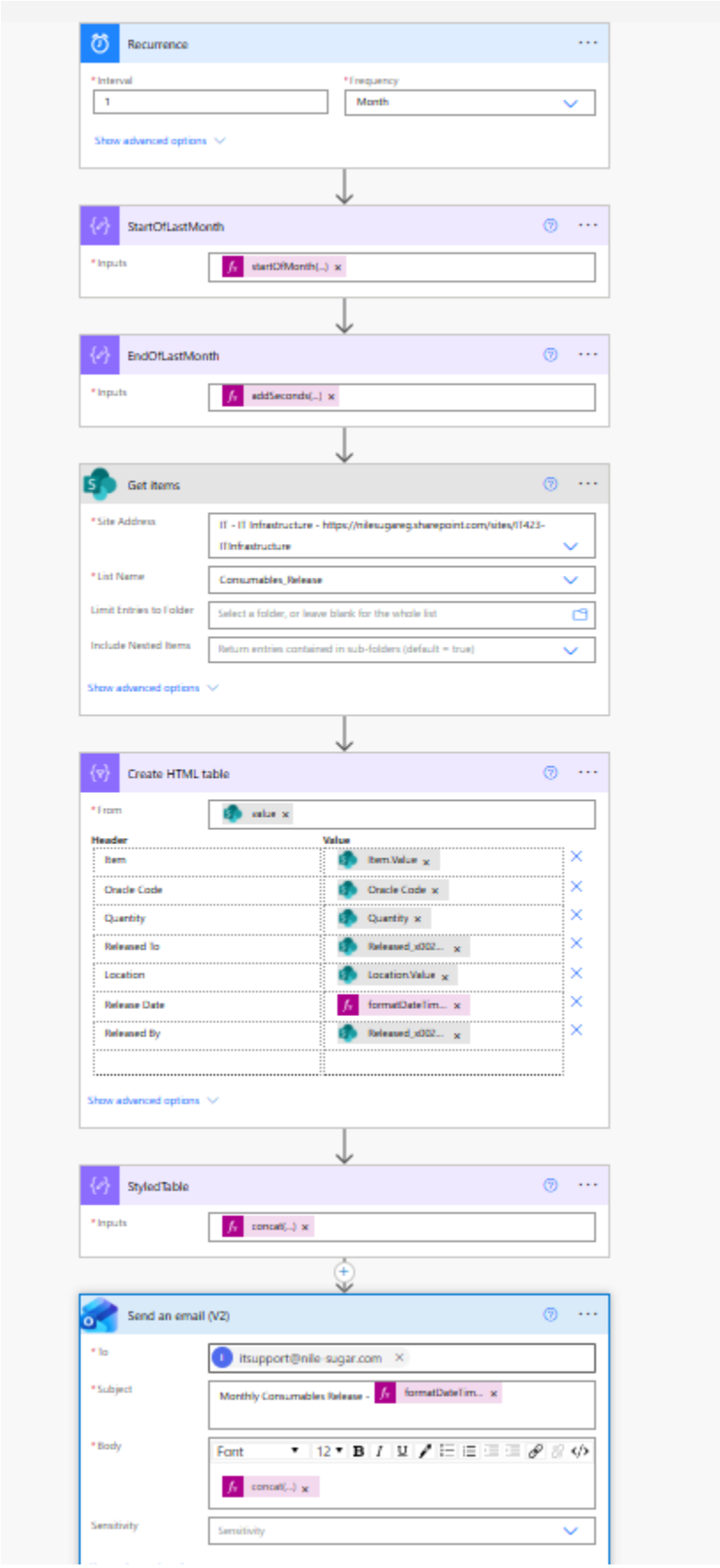
Please find below the monthly consumables stock status.

Item	Stock	No of Printers	To Order	Oracle Code	Site
HP 938 Yellow Cartridge [4S6X2A]	1	1	2	141001000118	FA
05A_CE505AC_HO	2	4	6	141001000033	HO
05A_CE505AC	6	1	-4	141001000033	FA
103a_w1103a	7	3	-1	141001000110	FA
106A_W1106A	6	9	12	141001000112	FA
124A_Q6000A	1	2	3	141001000054	FA
124A_Q6001A_Cyan	1	2	3	141001000053	FA
124A_Q6002A_Yellow	1	2	3	141001000052	FA
124A_Q6003A_Magenta	1	2	3	141001000051	FA
125A_CB540A _ Black	0	1	2	141001000057	FA
125A_CB541A_Cyan	0	1	2	141001000058	FA
125A_CB542A_Yellow	0	1	2	141001000059	FA
125A_CB543A_Magenta	0	1	2	141001000060	FA
12A_Q2612AC	5	1	-3	141001000028	FA
12A_Q2612AC_HO	5	3	1	141001000028	HO
130	0	1	2	141001000017	FA
134	0	1	2	141001000018	FA
14A_CF214XC	1	1	1	141001000078	FA

11:34 AM
1/21/2026

5. Monthly Release Report Email

Another scheduled flow runs monthly to extract all release transactions from the previous month. The data is filtered by date, formatted into a styled HTML table, and sent via email.



Monthly Release Report Email Template:

Monthly Consumables Release - January 2026

Power_Apps
To: Youssef Madian

Reply Reply all Forward
Wed 2026-01-21 08:47

Dears,

Please find below the monthly consumables release report for the previous month.

Period:
01/12/2025 to **31/12/2025**

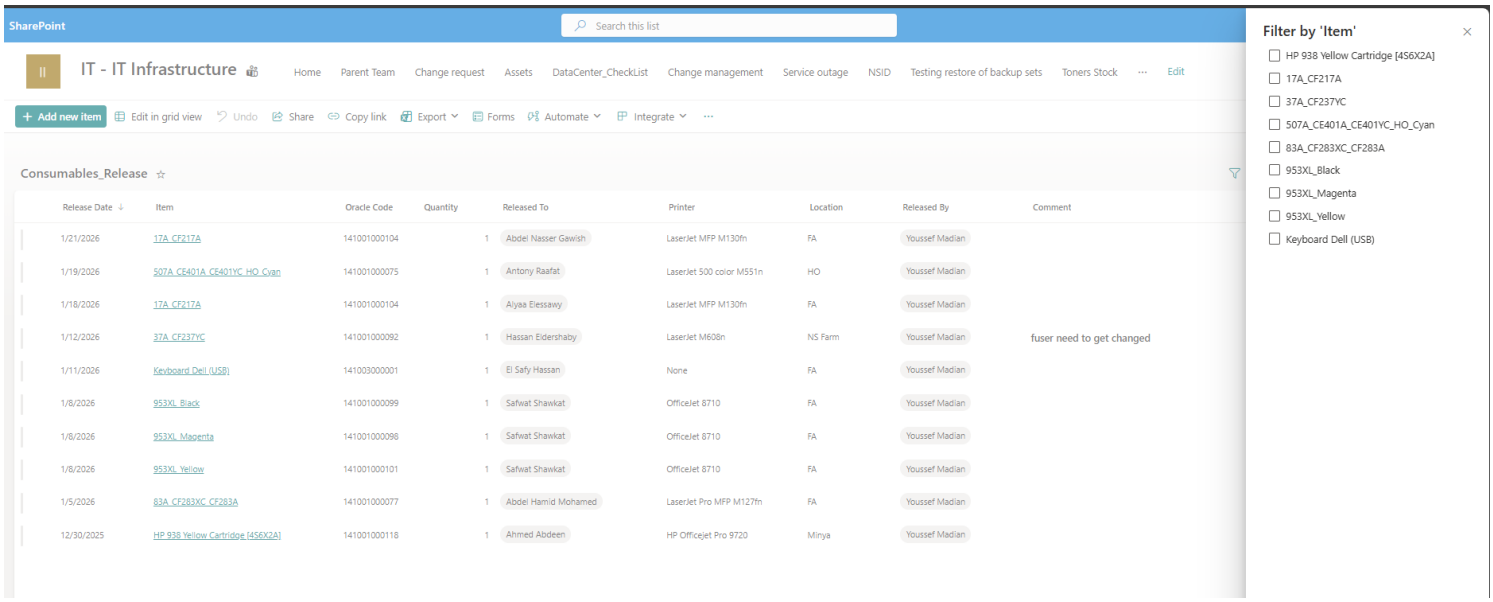
Item	Oracle Code	Quantity	Released To	Location	Release Date	Released By
HP 938 Yellow Cartridge [4S6X2A]	141001000118	3	Ahmed Abdeen	FA	03/12/2025	Youssef Madian
HP 938 Yellow Cartridge [4S6X2A]	141001000118	1	Albert Adib	HO	16/12/2025	Youssef Madian
HP 938 Yellow Cartridge [4S6X2A]	141001000118	3	Ahmed Areef	NS Farm	25/12/2025	Youssef Madian

Regards,
IT Infrastructure

Reply Forward

6. Audit & Traceability

All transactions remain stored in SharePoint, ensuring full auditability. Each release is traceable by date, user, location, and quantity.



Release Date	Item	Oracle Code	Quantity	Released To	Printer	Location	Released By	Comment
1/21/2026	17A_CF217A	141001000104	1	Abdel Nasser Gawish	LaserJet MFP M130fn	FA	Youssef Madian	
1/19/2026	507A_CE401A_CE401YC_HO_Cyan	141001000075	1	Antony Raafat	LaserJet 500 color M551n	HO	Youssef Madian	
1/18/2026	17A_CF217A	141001000104	1	Alyaa Elessawy	LaserJet MFP M130fn	FA	Youssef Madian	
1/12/2026	37A_CF237YC	141001000092	1	Hassan Eldershaby	LaserJet M608n	NS Farm	Youssef Madian	fuser need to get changed
1/11/2026	Keyboard Dell (USB)	141003000001	1	El Safy Hassan	None	FA	Youssef Madian	
1/8/2026	953XL_Black	141001000099	1	Safwat Shawkat	OfficeJet 8710	FA	Youssef Madian	
1/8/2026	953XL_Magenta	141001000098	1	Safwat Shawkat	OfficeJet 8710	FA	Youssef Madian	
1/8/2026	953XL_Yellow	141001000101	1	Safwat Shawkat	OfficeJet 8710	FA	Youssef Madian	
1/5/2026	83A_CF283XC_CF283A	141001000077	1	Abdel Hamid Mohamed	LaserJet Pro MFP M127fn	FA	Youssef Madian	
12/30/2025	HP 938 Yellow Cartridge [4S6X2A]	141001000118	1	Ahmed Abdeen	HP Officejet Pro 9720	Minya	Youssef Madian	